Q6

With batch gradient descent, the network considers the errors from all three examples before deciding how to adjust its weights. The resulting update reflects the combined gradient.

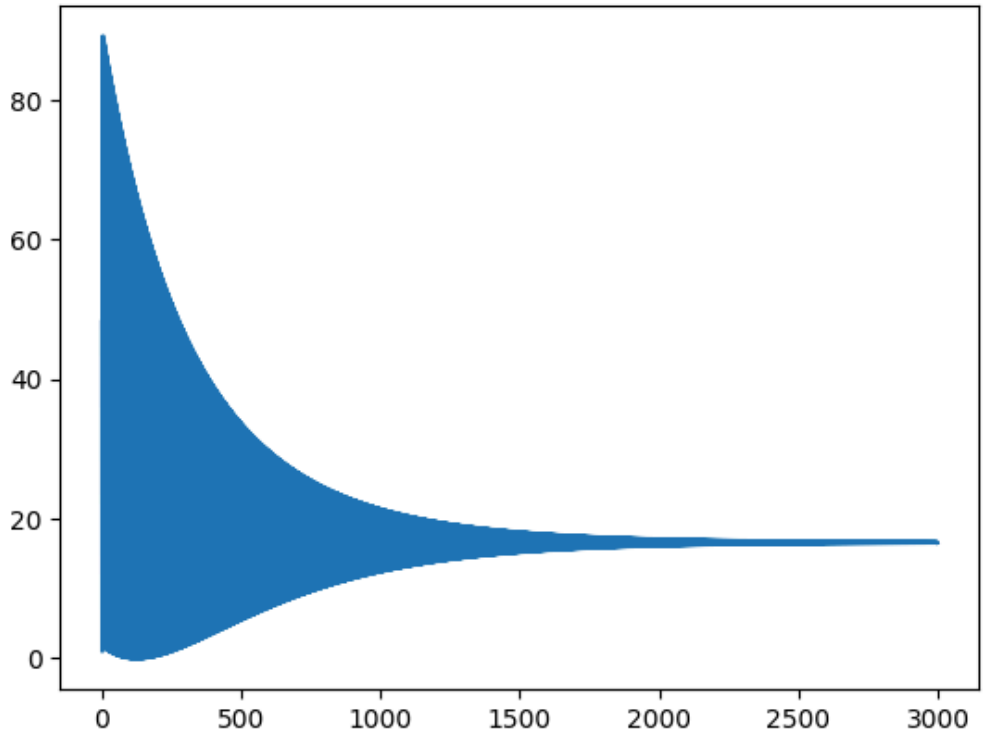
With SGD, the first example changes the weights. The second example then evaluates the model using the already updated weights and may push them in a different direction. The third example may push them in yet another direction.

Consequently, the loss can fluctuate rather than decrease smoothly.

Property	Batch gradient descent	SGD
Gradient calculation	Uses all three samples	Uses one sample at a time
Weight updates per epoch	1	3
Loss curve	Generally smoother	Generally noisier
Gradient estimate	Exact gradient of the full-batch loss	Individual-sample gradient
Computation per update	Higher	Lower
Weight changes during an epoch	Only after all samples	After every sample

Also, after 300 epochs, the loss remains high (envelope: 17.02-35.9). After running for 3000 epochs, the loss envelope decreases (16.01-16.04) but does not reach the low value of 5.47 observed in the Batch Gradient Descent case.

Stochastic gradient descent:



Batch Gradient Descent:

